Name ______________________________ Section ________

MTH:2310 DISCRETE MATH
QUIZ 4
DUE: FRIDAY OCT. 10

Complete all of the following problems being as detailed as you can be. You may work in groups, but you must submit your own solutions using your own words. You must submit a physical copy of your solutions to me unless told otherwise. If there is a specific number of problems which seem unsolvable let me know first, I may have made a typo. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit.

Handwriting: *(4 points)*
Your ability to express your solutions is just as important as mathematical accuracy.
You will be penalized up to 2 points for arithmetic errors.
You will be penalized up to 2 points for disorganized or hard to read solutions.

Problem 1. *(8 points) Sets*
Describe each of the following sets using roster notation, then determine the set's cardinality.

(a) *(4 points)* $A = \{x | (x \text{ is a U.S. state}) \wedge (x \text{ is in the Midwest})\}$

{ wisconsin, michigan, ohio, illinois, minnesota }

$|A| = 5$

(b) *(4 points)* $B = \{x | (x \in \mathbb{N}) \wedge (x \text{ is less than 20}) \wedge (x \text{ is divisible by } 3)\}$

$\{0, 3, 6, 9, 12, 15, 18\}$

$|B| = 7$

Problem 2. *(8 points) Sets*

Find the power set of $A = \Big\{1, 2, \Big\{3, \{1, 2\}\Big\}\Big\}$.

$P(A) = \{\{1\}, \{2\}, \{\{3, \{1,2\}\}\}, \{1, 2\}, \emptyset,$
$\{1, \{3, \{1,2\}\}\}, \{2, \{3, \{1,2\}\}\}, \{1, 2, \{3, \{1,2\}\}\}\}$

Problem 3. *(8 points) Sets*

Determine whether these statements are true or false.

(a) *(2 points)* $\emptyset \in \{\emptyset\}$

True

(b) *(2 points)* $\emptyset \subseteq \{\emptyset\}$

True

(c) *(2 points)*
$\{\emptyset\} \in \{\emptyset, \{\emptyset, \emptyset\}, \emptyset\}$

True

$\{\emptyset\} = \{\emptyset, \emptyset\}$

(d) *(2 points)*
$\{\{\emptyset\}, \emptyset\} \subseteq \{\emptyset, \{\emptyset, \emptyset\}, \emptyset\}$

True

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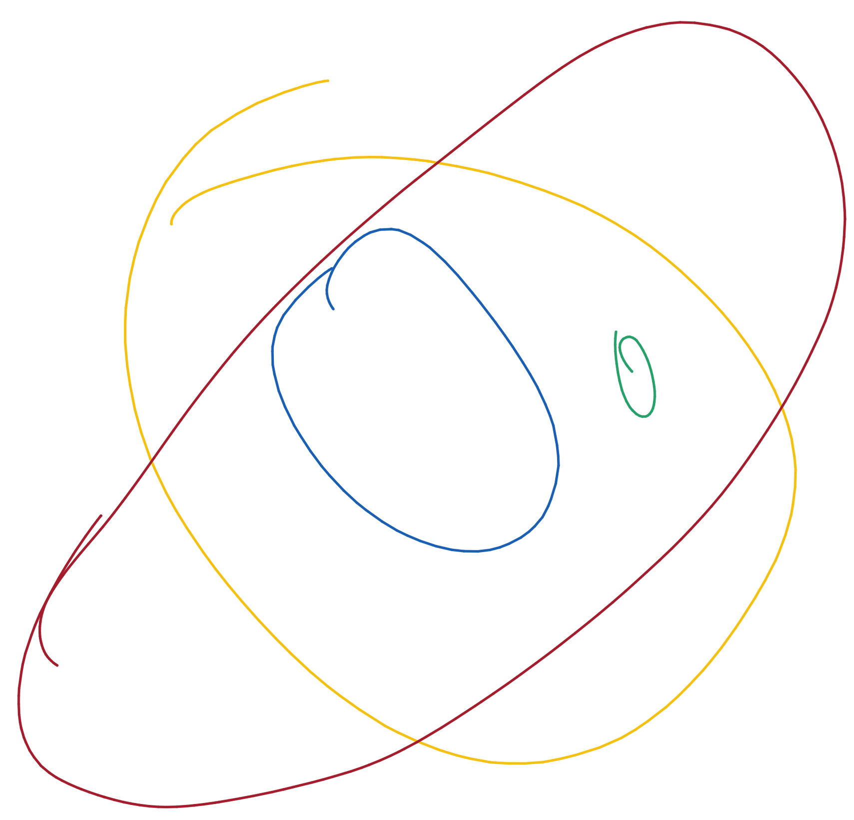
Problem 4. *(14 points) Set Operations*

Let U be the universal set. Let A, B, C, and D be distinct non-empty sets. Suppose that

A, B, C, D

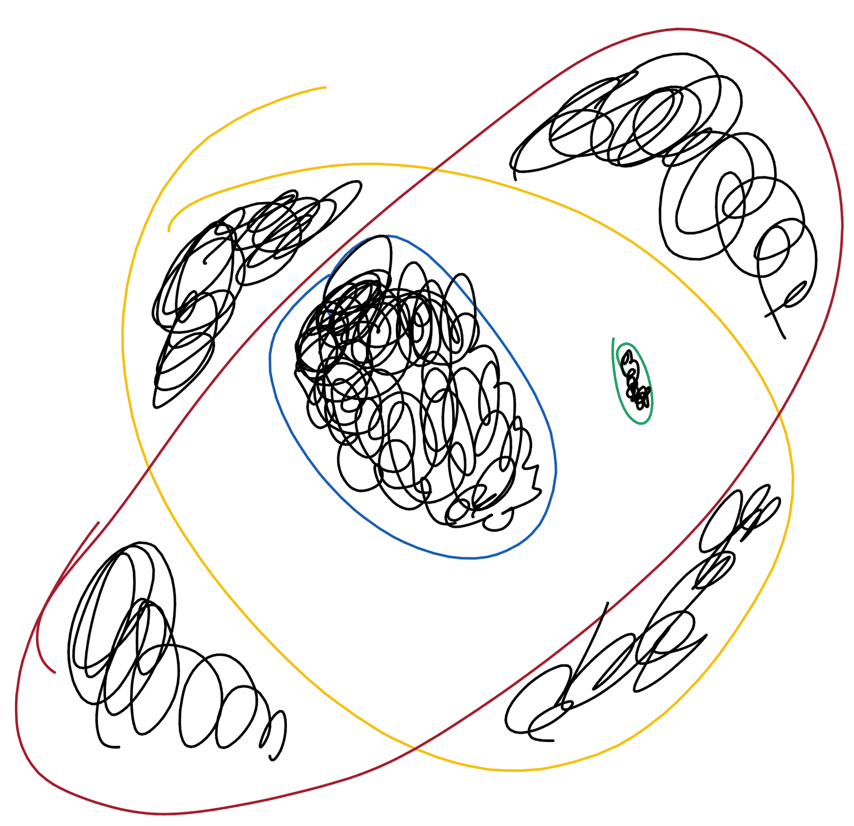
- $A \subset C$
- $B \subset C$
- $C \not\subset D$
- $A \cap B = \emptyset$
- $A \subset D$
- $B \subset D$
- $D \not\subset C$

(a) *(10 points)* Use a Venn diagram to illustrate all of these relationships.



(b) *(4 points)* Redraw the diagram you made for part (a) and shade the region corresponding to

$$\Big(\big(C \cup D\big) \setminus \big(C \cap D\big)\Big) \cup \big(A \cup B\big)$$



Problem 5. *(8 points) Set Operations*
Let A, B, and C be non-empty sets. Show that $\Big(B \setminus A\Big) \cup \Big(C \setminus A\Big) = \Big(B \cup C\Big) \setminus A$ using set-builder notation only.

$$(B \setminus A) \cup (C \setminus A) = \{x \mid (x \in B \wedge x \notin A) \vee (x \in C \wedge x \notin A)\}$$

$$(B \cup C) \setminus A = \{x \mid (x \in B \vee x \in C) \wedge x \in A$$

$$(x \in B \vee x \in C) \wedge x \in A = (x \in B \wedge x \notin A) \vee (x \in C \wedge x \notin A)$$

therefore $(B \setminus A) \cup (C \setminus A) = (B \cup C) \setminus A$

Problem 6. *(8 points) Functions*
Use the definition for an injective function to show that the function $f : \mathbb{R} \to \mathbb{R}$ is injective where f is defined as $f(x) = 2x^3 - 1$

injectivity requires for all $\mathbb{R}$ $f(a) = f(b)$ where $a = b$

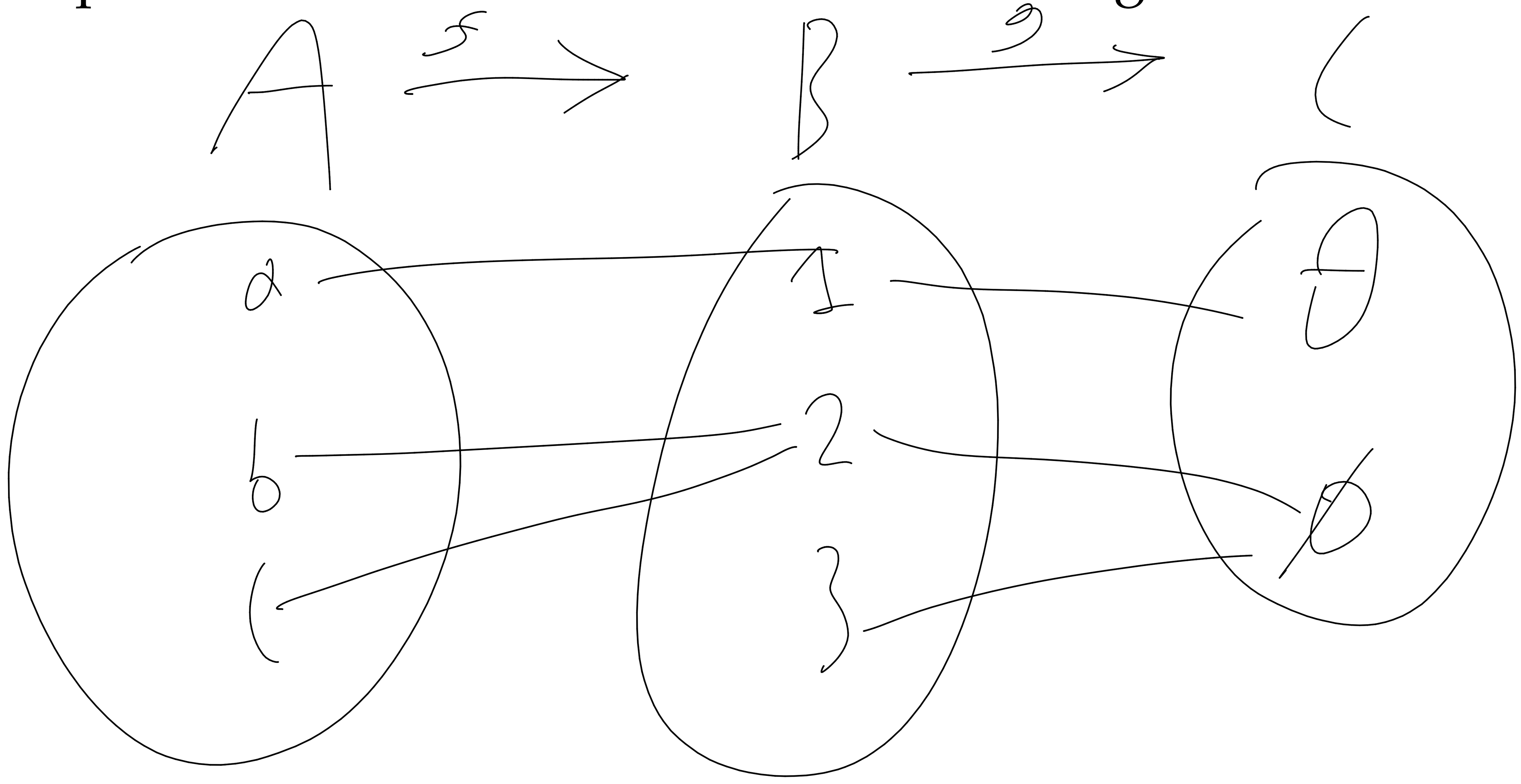
Suppose $f(a) = f(b)$

$$2a^3 - 1 = 2b^3 - 1$$
$$+1 \qquad +1$$
$$\overline{\qquad 2 \qquad}$$
$$a^3 = b^3$$
$$a = b$$

thus as $a = b$ when $f(a) = f(b)$,
we show that $f : \mathbb{R} \to \mathbb{R}$
is injective where $f(x) = 2x^3 - 1$

Problem 7. *(12 points) Functions*
Let $f : A \to B$, $g : B \to C$, and $h : C \to A$ be functions where $A = \{a, b, c\}$, $B = \{1, 2, 3\}$, and $C = \{\theta, \phi\}$.

(a) *(6 points)* Define f and g in such a way that the composition $g \circ f$ is surjective, but ~~g~~ f is not surjective. You may use an explicit definition or a "bubble-arrow" diagram.



(b) *(6 points)* Define h and f in such a way that the composition $f \circ h$ is injective, but f is not injective. You may use an explicit definition or a "bubble-arrow" diagram.